

LearnForge Support Assistant

Data Schema and Indexed Representation

Purpose. This document specifies the data model used to convert the supplied LearnForge support corpus into Qdrant records. The corpus contains 40 logical records: 15 FAQs, 10 policies, and 15 historical support tickets. A logical source entry is retained as one record so that policy conditions, provenance, and citations remain readable.

1. Canonical knowledge record

Every source entry is parsed into a validated **KnowledgeRecord**. The complete source text is retained; the additional fields support retrieval, provenance, and safe conflict resolution.

Field	Type	Meaning
<code>record_id</code>	string	Stable source identifier, for example <code>FAQ-07</code> , <code>POLICY-02</code> , or <code>TICKET-04</code> .
<code>source_type</code>	enum	<code>faq</code> , <code>policy</code> , or <code>ticket</code> .
<code>title</code>	string	Source heading or primary subject.
<code>text</code>	string	Complete, unmodified source entry.
<code>source_date</code>	ISO date / null	Explicit effective, reviewed, or updated date when stated by the source.
<code>temporal_status</code>	enum	<code>current</code> , <code>current_unversioned</code> , or <code>historical</code> .
<code>authority_tier</code>	enum	<code>policy</code> , <code>faq</code> , or <code>historical_example</code> .
<code>contains_deprecated_reference</code>	boolean	Indicates that the record mentions retired or superseded guidance.
<code>ticket_status</code>	string / null	Resolution state, populated only for support tickets.
<code>content_hash</code>	SHA-256 string	Hash of canonical record content and metadata for change detection.

2. Qdrant point design

Each canonical record is stored as one Qdrant point in the `learnforge_support` collection. Point IDs are deterministic UUIDv5 values derived from `record_id`; rebuilding the index is therefore idempotent.

Point component	Implementation	Purpose
Point ID	UUIDv5 of <code>record_id</code>	Stable identity across rebuilds.
dense vector	BAAI/bge-small-en-v1.5; 384 dimensions; cosine distance	Semantic similarity retrieval for paraphrases and conversational wording.
bm25 vector	Qdrant/bm25; sparse indices and values; IDF modifier	Exact-term retrieval for policy IDs, durations, product names, and billing language.
Payload	Serialized KnowledgeRecord	Preserves source text, metadata, citations, and reliability context.

The text indexed into both vector representations concatenates the title, two newline separators, and the complete source text. Embeddings are retrieval representations only; they are not treated as source truth or exposed in application traces.

3. Retrieval and reliability fields

At request time, dense and BM25 searches each return candidate records. Reciprocal Rank Fusion combines the ranked lists, after which a cross-encoder reranks the shortlist. The following fields exist only during a request and are not written into the canonical payload:

- **dense_rank** and **sparse_rank**: 1-indexed positions in each first-stage result list.
- **rrf_score**: fused ranking score; raw dense and BM25 scores are never directly combined.
- **reranker_score**: relevance score produced by the cross-encoder.

Evidence is ordered by authority and freshness before generation:

current policy > FAQ > historical support ticket

Historical tickets can illustrate a previous resolution but cannot establish current policy. When evidence is insufficient, conflicting, or requires an account action, the system returns a clarification or escalation outcome instead of inventing a policy answer.

4. Example payload

```
{
  "record_id": "POLICY-02",
  "source_type": "policy",
  "title": "Cancellation and Refund Policy",
  "source_date": "2026-01-01",
  "temporal_status": "current",
  "authority_tier": "policy",
  "contains_deprecated_reference": true,
  "ticket_status": null,
  "content_hash": "<sha256>",
  "text": "<complete source entry>"
}
```